VI SEMESTER	L	T	P	C
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20AM6T02 - CLOUD ARCHITECTURE AND SERVICES				

COURSE OUTCOMES:
After completion of course, students would be able to:
CO1: Understanding Architecture and components of cloud computing
CO2: Understanding the key dimensions of the challenge of Cloud Computing
CO3: Assessment of the economics , financial, and technological implications for selecting cloud computing for own organization
CO4: Assessing the financial, technological, and organizational capacity of employer's for actively initiating and installing cloud-based applications.
CO5: Understanding the Cloud applications.

UNIT 1:
Cloud Platform Architecture: Introduction to Cloud computing, Cloud Computing Architecture, Cloud computing services ,Infrastructure as a service(Iaas), Platform as a service (Paas), Software as a service (Saas), Cloud computing Deployment models or types of cloud ,Public cloud, Private cloud, Hybrid cloud , Advantages of Cloud computing, Service Oriented Architecture (SOA), Cloud infrastructure, Economics of the cloud

UNIT-II:
Virtualization :Introduction, Characteristics of virtual environment, Classification of Virtual techniques, Virtualization and Cloud computing, Pros and cons of Virtualization ,Technology examples-VMware and Microsoft Hyper-V.

UNIT-III:
Introduction to Cloud Platforms: Amazon Web Services (AWS), Global Infrastructure, AWS services: compute, storage, database, networking, Introduction to Microsoft Azure, Introduction to Google App Engine (GAE).

UNIT-IV:
Cloud Security: Security, Security planning, understanding security of cloud, understanding data security, Encryption, cloud computing applications.
Software as a Service Security, Virtual machine security, Identity and access management

UNIT-V:
Cloud Applications: Scientific Applications, Health care, Geo science and Biology, Business and consumer applications, CRM and ERP, Social networking, Media Applications
Storage Systems: Evolution of storage technology, storage models, files systems and database, Amazon Simple Storage Service (S3).

Text Books:
1. Distributed and Cloud Computing, Kai Hwang, Geoffry C. Fox, Jack J. Dongarra MK Elsevier.
2. Mastering Cloud Computing, Foundations and Application Programming, Raj Kumar Buyya, Christen vecctiola, S Tammaraiselvi, TMH.
3. Cloud Computing, A Hands on approach, ArshadeepBahga, Vijay Madisetti, University Press.
4. Cloud computing, Master cloud computing concepts,architecture and applications with real world examples and case studies, Kamal kanthiran,RuchiDoshi

Reference Books:

1. Cloud Computing, A Practical Approach, Anthony T Velte, Toby J Velte, Robert Elsenpeter, TMH.
2. Raj Kumar Buyya,"Cloud Computing: Principles and Paradigms",Wiley,2011.
3. Cloud Computing, Theory and Practice, Dan C Marinescu, MK Elsevier.

Links:

https://aws.amazon.com/what-is-cloud-computing/
https://www.geeksforgeeks.org/virtualization-cloud-computing-types/?ref=lbp
http://www.itbriefcase.net/network-server-storage-and-desktop-virtualization
https://www.javatpoint.com/virtualization-in-cloud-computing
https://www.tutorialspoint.com/cloud_computing/cloud_computing_security.htm